



Drug used in the treatment of gout

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Objectives

By the end of this lecture, students should be able to:

- **Define gout and explain its clinical significance.**
- **Classify the different types of gout.**
- **Explain the pathophysiological mechanisms underlying gout.**
- **Describe the pharmacological management of gout.**
- **Identify the major adverse effects associated with commonly used anti-gout medications.**

Definition

- It is painful joint inflammation, most commonly in the first metatarsophalangeal joint, resulting from precipitation of monosodium urate crystals in a joint space.
- Gout is typically diagnosed by identification of monosodium urate crystals in synovial fluid of the affected joint.





Types of Gout

Etiology-based

Primary gout: genetic and metabolic.

Secondary gout: due to other diseases or drugs

Clinical course-based

Acute gout

Chronic tophaceous gout

Causes of Gout

- Primary Gout

90% of cases stems from genetic factors affecting uric acid metabolism or renal handling.

(Renal Underexcretion, Overproduction of uric acid by the body and inherited predisposition to metabolic dysfunction or transporter defects, such as *SLC2A9* (GLUT9))

- Secondary Gout

Increased uric acid production (e.g., tumor lysis syndrome, myeloproliferative disorders)

Decreased excretion (e.g., chronic kidney disease)

Drugs (e.g., diuretics)

The Stages of Gout Progression

STAGE 1: High Uric Acid Levels

Uric acid is building up in the blood and starting to form crystals around joints

STAGE 2: Acute Gout

Symptoms start to occur, causing a painful gout attack

STAGE 3: Intercritical Gout

Periods of remission between gout attacks

STAGE 4: Chronic Gout

Gout pain is frequent and tophi form in joints



Risk Factors of Gout



Male > Female



Food Rich
in Fructose



Excessive
Alcohol Intake



Obesity



Diet Rich
in Purines



Chronic Kidney
Disease

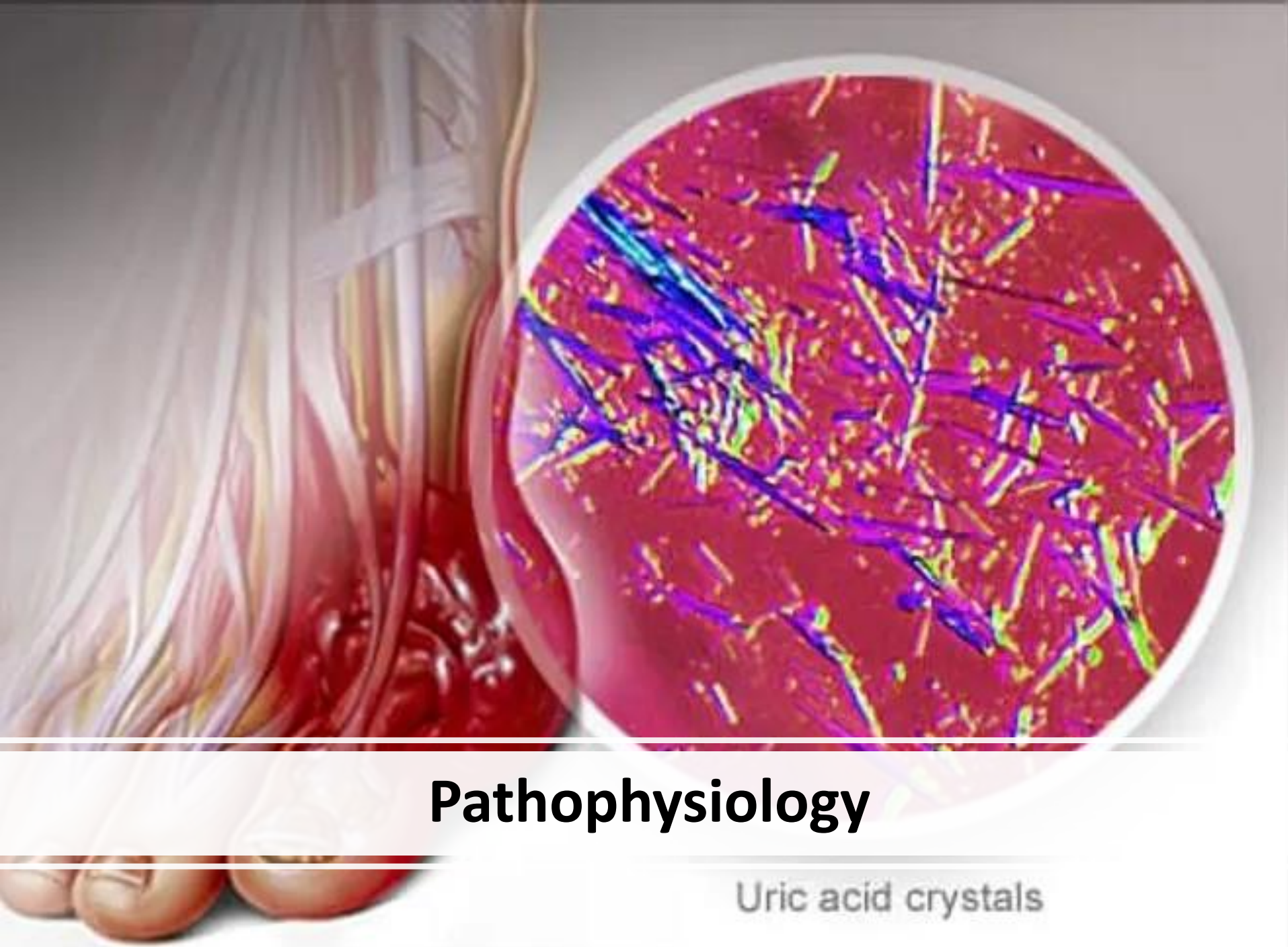
Symptoms of Gout





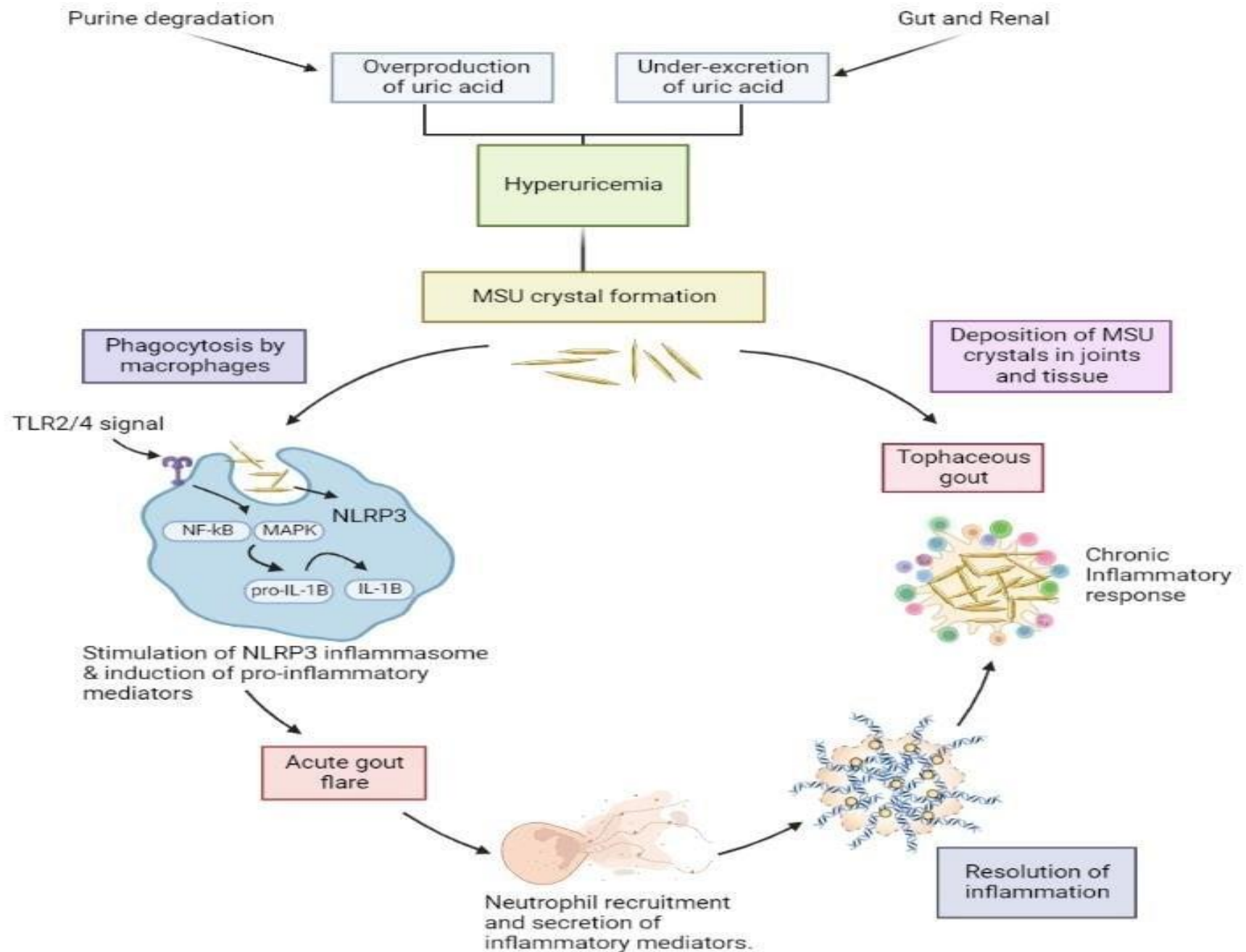
Tophi

**subcutaneous nodules
comprised of
monosodium urate
crystals in a matrix of
lipids, protein, and
mucopolysaccharides)
may also form at the
joint space**



Pathophysiology

Uric acid crystals



The image features a vibrant blue background with several blister packs and loose pills. In the upper left, there are blister packs containing blue, oval-shaped capsules. Below these, there are several blister packs containing white, round tablets, some of which have a score line. A few white tablets are also scattered on the blue surface. The overall composition is clean and modern, with a focus on pharmaceuticals.

Management

Treatment

**Acute
management**

**Chronic
management**

**Lifestyle
modification**

Acute gout treatment

A. NSAIDs (*First-line*)

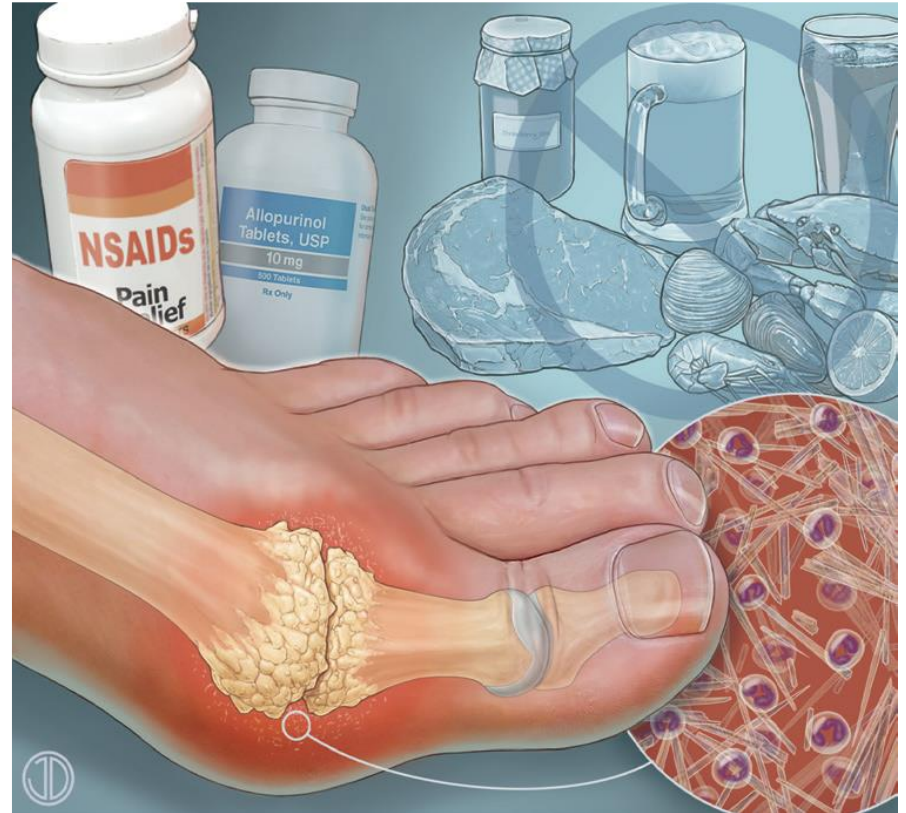
Examples: Indomethacin,
Naproxen

Mechanism: Inhibit prostaglandin
synthesis (↓ inflammation)

Use: Start early in attack

**Any oral NSAID may be given at the
maximal dosage and continued for
one to two days after relief of
symptoms.**

Side effects : Gastric irritation,
peptic ulcer and renal impairment



B. Colchicine

- **Mechanism:** Inhibits microtubule formation →
↓ neutrophil migration
- **Best if given within first 24 hours**
- **Side effects :** Diarrhea, Nausea/vomiting, and Bone marrow suppression



C. Corticosteroids

- **Examples:** Prednisone (oral), intra-articular steroids
- **Used when** NSAIDs/colchicine are contraindicated
- **When gout is limited to a single joint,** intra-articular corticosteroid injections may be preferable to systemic corticosteroids because of their lower adverse effect profile.
- **Rebound flares** are common after discontinuation of corticosteroid. To reduce the risk preventive treatment and initiation of a tapered course of corticosteroids over 10 to 14 days is recommended after resolution of symptoms.



Acute gout treatment

Medication	Example regimen	Notes
NSAIDs	Indomethacin (Indocin), 50 mg three times per day	First-line therapy; all NSAIDs are equally effective; adverse effects include gastric bleeding and kidney injury
Colchicine (Colcrys)	1.2 mg initially, then 0.6 mg one hour later, then 0.6 to 1.2 mg per day	No analgesic properties; gastrointestinal adverse effects are common; avoid use in patients with renal and hepatic insufficiency; contraindicated in patients receiving clarithromycin (Biaxin)
Corticosteroids	Oral, intramuscular, or intra-articular routes, variable dosing (e.g., prednisone, 40 mg for four days, then 20 mg for four days, then 10 mg for four days)	Preferred therapy for patients in whom NSAIDs and colchicine are contraindicated; when discontinuing oral corticosteroids, taper to avoid rebound flares

Chronic (Long-Term) Management

- **Goal:**

1. Long-term urate lowering therapy (ULT) to reduce serum uric acid levels below 6 mg/dL,
2. Prevent future flares and dissolve existing crystals (tophi).

ULT should be continued for three to six months after a flare if there are no ongoing symptoms. Therapy should continue indefinitely if there are ongoing signs or symptoms (e.g., one or more tophi on examination)



Chronic Management



**Lifestyle and
Dietary
modification**



**Pharmacologic
treatment**

Lifestyle and Dietary modification

- **Dietary Adjustments:** Limit consumption of high-purine foods, such as red meat, organ meats, and shellfish.
- **Beverages:** Avoid alcohol and high-fructose corn syrup.
- **Hydration:** Drink plenty of water to help kidneys eliminate uric acid.
- **Weight Management:** Gradual weight loss can reduce uric acid levels.



Pharmacologic treatment

- **Urate-Lowering Therapy (ULT):**

Target serum uric acid **< 6 mg/dL** (or **< 5 mg/dL in severe cases**) to prevent crystal deposition and reduce tophi.

- **Prophylaxis:**

Colchicine (0.5–1 mg/day) or low-dose NSAIDs for **3–6 months** to prevent flares when starting ULT.

- **Refractory Cases:**

Pegloticase (IV) for severe, treatment-resistant gout.



Urate-Lowering Drugs

Xanthine Oxidase Inhibitors (XOIs):

Allopurinol, Febuxostat

→ Decrease uric acid production

Uricosuric Agents:

Probenecid

→ Increase renal excretion of uric acid

Uricase Agents:

Pegloticase

→ Convert uric acid into allantoin (more soluble)

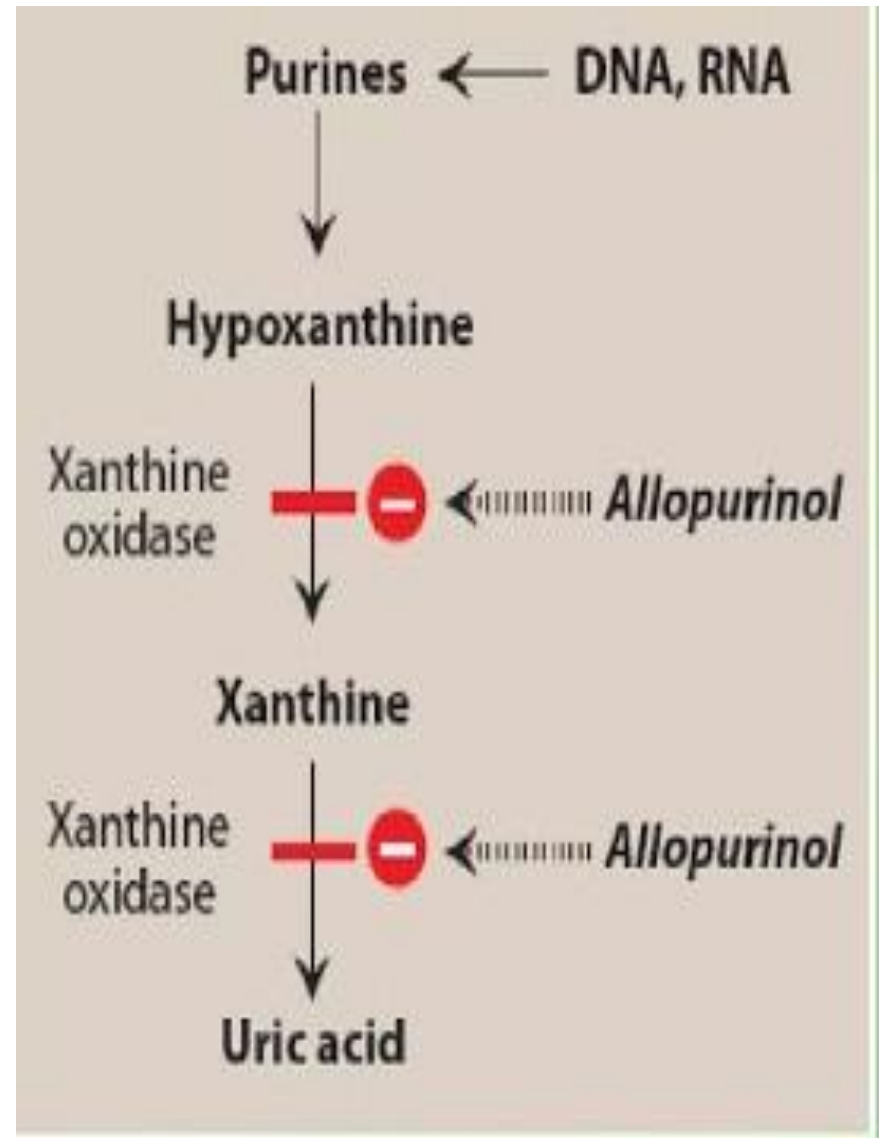
Xanthine Oxidase Inhibitors

- **Drugs:** Allopurinol, Febuxostat
- **Mechanism of Action**

Inhibit xanthine oxidase enzyme

↓ Conversion of hypoxanthine & xanthine → uric acid

Result: ↓ serum uric acid levels



- Indications

Chronic gout (first-line therapy)

Prevention of gout attacks

Tumor lysis syndrome (especially allopurinol)

- Adverse Effects

Allopurinol:

- Rash (can be severe → Stevens–Johnson syndrome)
- GI upset
- Hepatotoxicity

Febuxostat:

- Liver function abnormalities
- ↑ Risk of cardiovascular events (important point)

- Use Febuxostat if:

Patient cannot tolerate allopurinol

Or allopurinol is ineffective

Probenecid (Uricosuric Agent)



- Mechanism of Action

Inhibits URAT1 transporter in renal tubules

↓ Reabsorption of uric acid → ↑
uric acid excretion in urine

- Adverse Effects

Uric acid kidney stones

GI upset

Rash

Uricase Agents (Pegloticase)

- Mechanism of Action

Recombinant **uricase enzyme**

Converts **uric acid** → **allantoin** (more soluble, easily excreted)

- Indications

Severe chronic gout (refractory cases)

Patients who:

- Do not respond to allopurinol/febuxostat
- Cannot tolerate standard therapy



- **Route & Administration**

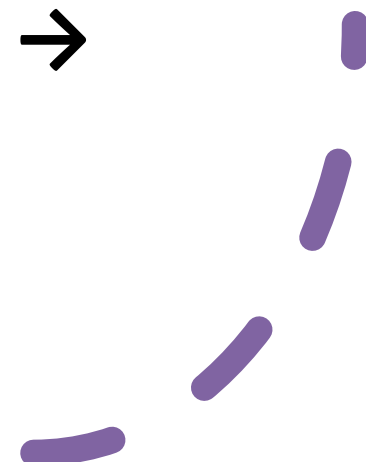
Given intravenously (IV infusion)

Usually every 2 weeks

- **Adverse Effects**

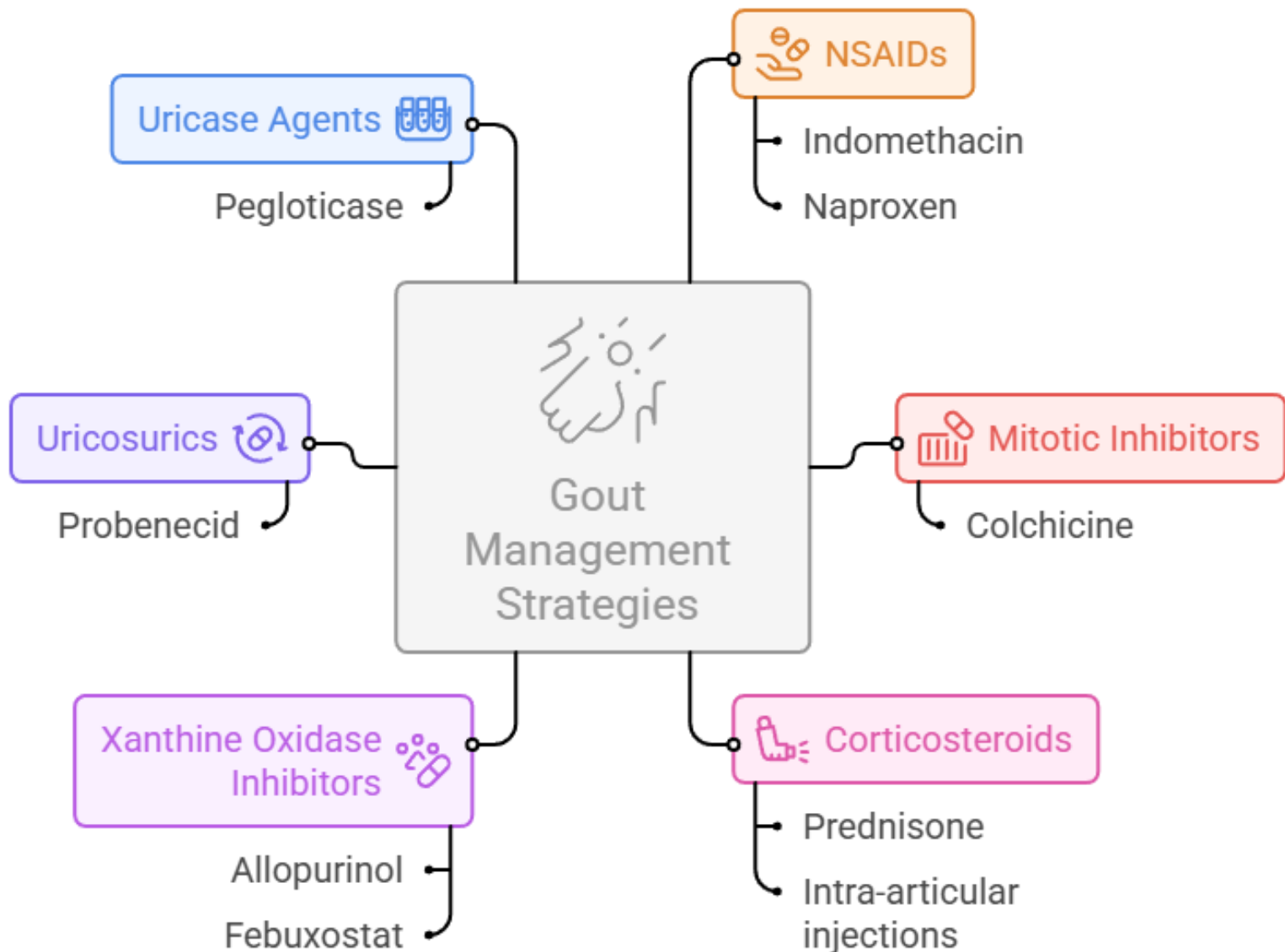
Infusion reactions (common)

Anaphylaxis (serious) → requires monitoring



<i>Medication</i>	<i>Dosage</i>	<i>Notes</i>
Colchicine (Colcrys)	0.6 to 1.2 mg per day	May cause reversible axonal neuromyopathy; may increase risk of rhabdomyolysis when used with statins or clarithromycin (Biaxin)
Pegloticase (Krystexxa)	8 mg intravenously every two weeks	Indicated for refractory gout; expensive (more than \$5,000 per dose ²⁸)
Probenecid	250 mg two times per day initially; titrate up to 2 g per day	High risk of nephrolithiasis; encourage hydration and urine alkalization with potassium citrate; multiple drug interactions
Xanthine oxidase inhibitors:		
Allopurinol (Zyloprim)	100 mg per day initially, except in patients with renal dysfunction; common effective dosage is 300 mg per day, but higher dosages may be needed	Genetic testing recommended before initiating treatment in patients at risk of severe hypersensitivity skin reaction (those of Han Chinese or Thai descent, regardless of kidney function, or Koreans with chronic kidney disease stage 3 or greater)
Febuxostat (Uloric)	40 mg once per day; may increase up to 80 mg per day if serum uric acid level > 6 mg per dL (357 μ mol per L) after two weeks	Contraindicated in patients receiving azathioprine (Imuran) and mercaptopurine

Gout Management Strategies



Thank you



D. Interleukin-1 (IL-1) Inhibitors

- **Examples:** Anakinra (off-label use), Canakinumab (FDA-approved)
- **Mechanism of Action:**
- Directly neutralizes Interleukin-1 (IL-1), halting the severe inflammatory cascade triggered by the NLRP3 inflammasome.
- **Indications:**
 - Used for acute gout flares when standard therapies (NSAIDs, colchicine, corticosteroids) are **contraindicated**, ineffective, or poorly tolerated.
 - Particularly useful for patients with severe chronic kidney disease (CKD), poorly controlled diabetes, or active gastrointestinal bleeding.
- **Route of Administration:** Subcutaneous (SC) injection.
- **Adverse Effects & Limitations:**
 - Increased risk of infections.
 - Injection site reactions.
 - High cost compared to first-line therapies.